



Hemoglobinopathies

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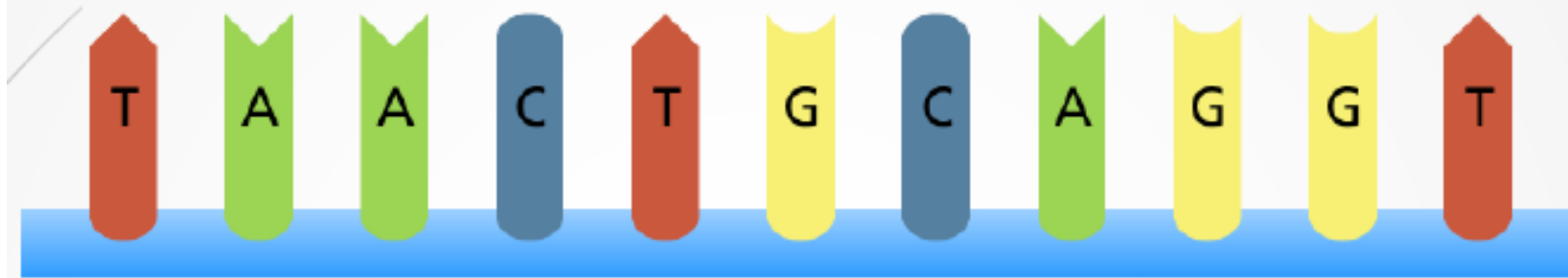
- Hemoglobinopathy refers to a disease state (opathy) involving the hemoglobin (Hb) molecule.
- Hemoglobinopathies are the most common genetic diseases, affecting approximately 7% of the world's population.
- Approximately 300,000 children are born each year with some form of inherited hemoglobin disorder.
- All hemoglobinopathies result from a genetic mutation in one or more genes that affect hemoglobin synthesis.

- In qualitative hemoglobinopathies, hemoglobin synthesis occurs at a normal or near-normal rate, but the hemoglobin molecule has an altered amino acid sequence within the globin chains.
- This change in amino acid sequence alters the structure of the hemoglobin molecule (structural defect) and its function (qualitative defect).
- Are a group of genetic or inherited diseases characterized by defects in the structure of globin chains of hemoglobin (qualitative or structural abnormalities).

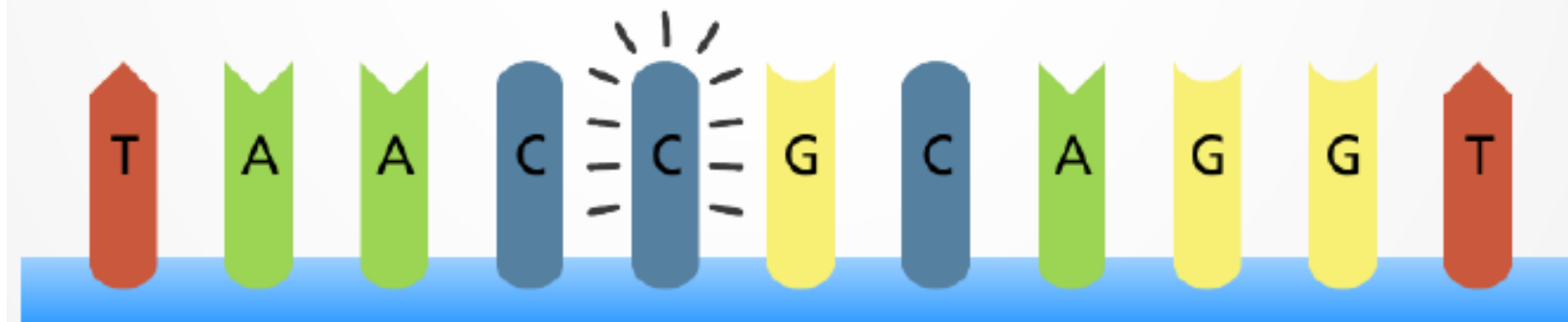
- More than 1000 structural hemoglobin variants (Hemoglobinopathies) are known to exist throughout the world. However, the majority of these variants are not clinically significant.
- The types of genetic mutations that occur in Hemoglobinopathies include point mutations, deletions, insertions, chain extension, and fusions involving one or more of the adult globin genes (α , β , γ , and δ).
- Point mutation is the most common type of genetic mutation occurring in the Hemoglobinopathies.

- Point mutation is the replacement of one original nucleotide in the normal gene with a different nucleotide, and the reading frame remains intact.
- This results in the substitution of one amino acid in the globin chain product at the position corresponding to the location of the original point mutation.
- More than 90% of the hemoglobin variants are single amino acid substitutions in the α , β , δ , and γ globin chains as a result of a single-point mutation in one of the globin chains.
- 1109 known Hemoglobinopathies variants result from a point mutation that causes an amino acid substitution.
- Hemoglobinopathies are inherited as a codominant trait.

Original sequence



Point mutation



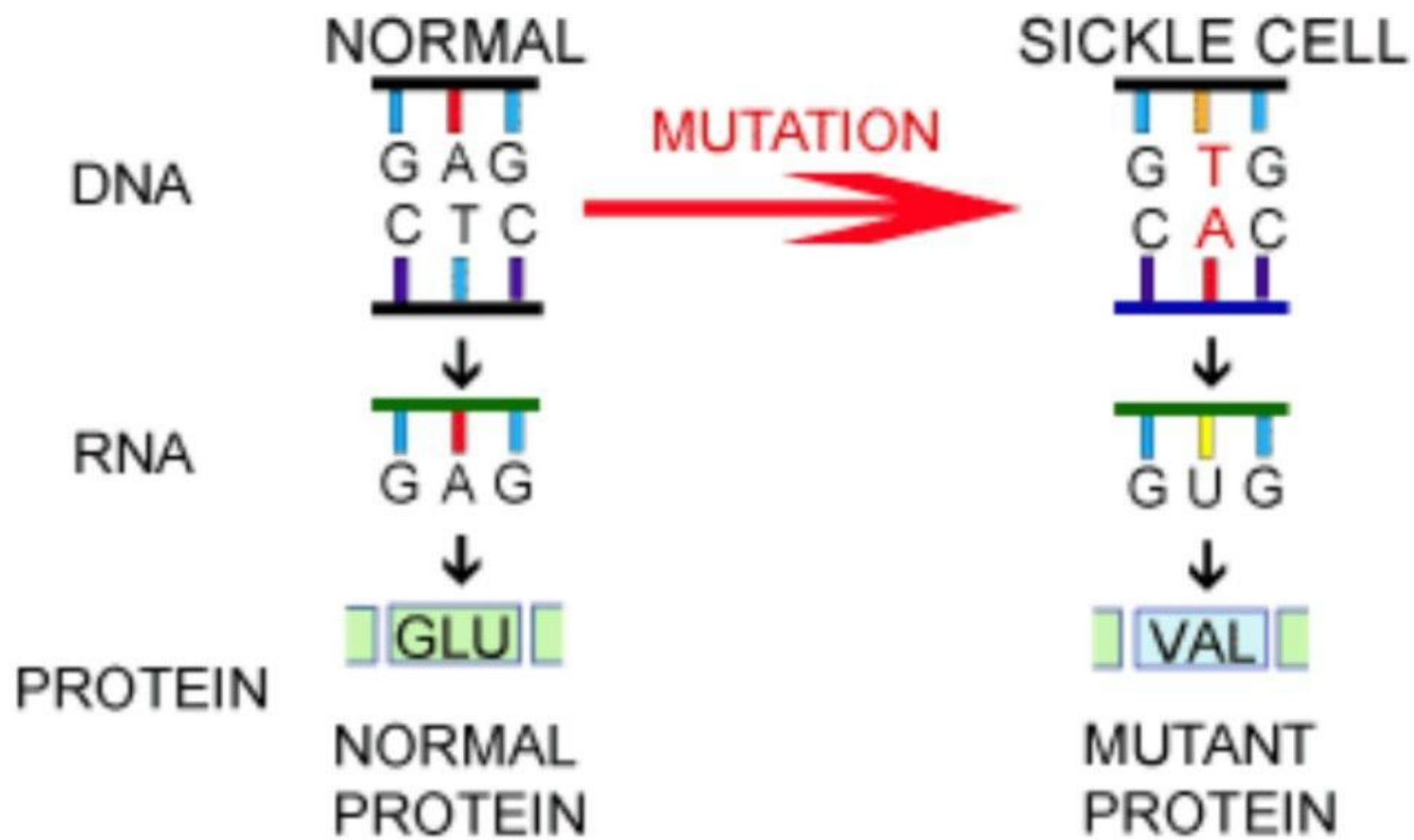
- Changes in Hb structure usually affect its function.
- Most of the Hb variants are present in the β chain followed by the α chain.
- There are numerous structural Hbs variants, of which the abnormal Hbs that result in hemolytic anemia is one of the most clinically significant variants.
- The most frequently occurring of the abnormal Hbs and the most severe is Hb S.

The most common variants of Hb

- Hb S: $\alpha_2 \beta_2$ $6\text{Glu} \rightarrow \text{Val}$, the 6th amino acid in the β -chain, glutamic acid, is substituted by valine.
- Hb C: $\alpha_2 \beta_2$ $6\text{Glu} \rightarrow \text{Lys}$, the 6th amino acid in the β -chain, glutamic acid, is substituted by lysine.
- Hb D: $\alpha_2 \beta_2$ $121\text{Glu} \rightarrow \text{Gln}$, the 121st amino acid in the β -chain, glutamic acid, is substituted by glutamine.
- Hb E: $\alpha_2 \beta_2$ $26\text{Glu} \rightarrow \text{Lys}$, the 26th amino acid in the β -chain, glutamic acid, is substituted by lysine.

Hemoglobin S, Sickle cell anemia

- It is the most common and the most severe abnormal Hb.
- It is caused by a point mutation, in which the 6th amino acid in the β -chain, glutamic acid, is substituted by valine (The affected chain is the β chain).
- $A_2 \beta_2^{6\text{Glu} \rightarrow \text{Val}}$
- There are two forms of HbS:
 - 1) The heterozygous or trait (HbAS): one β chain is mutated and the other is normal.
 - 2) The homozygous or disease (HbSS): both β chains are mutated.

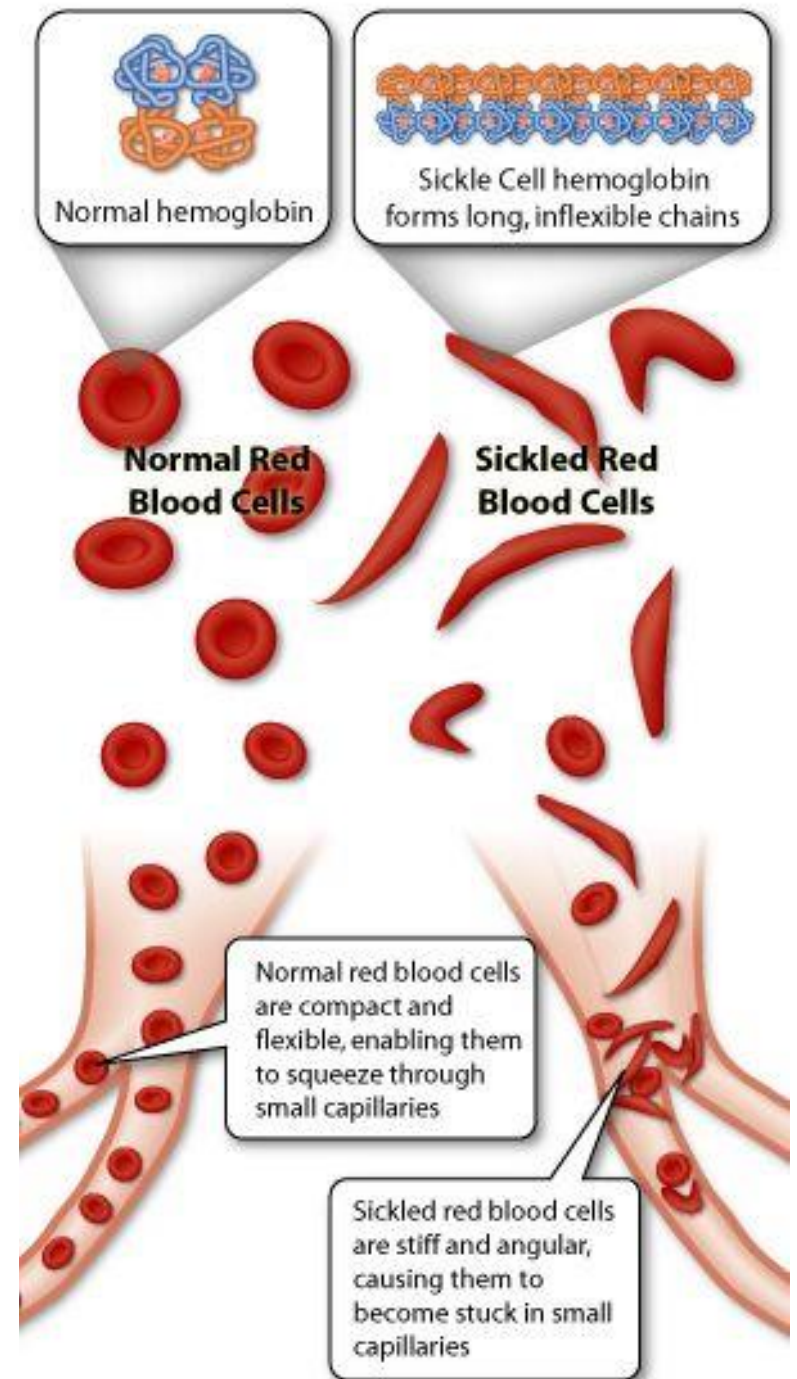
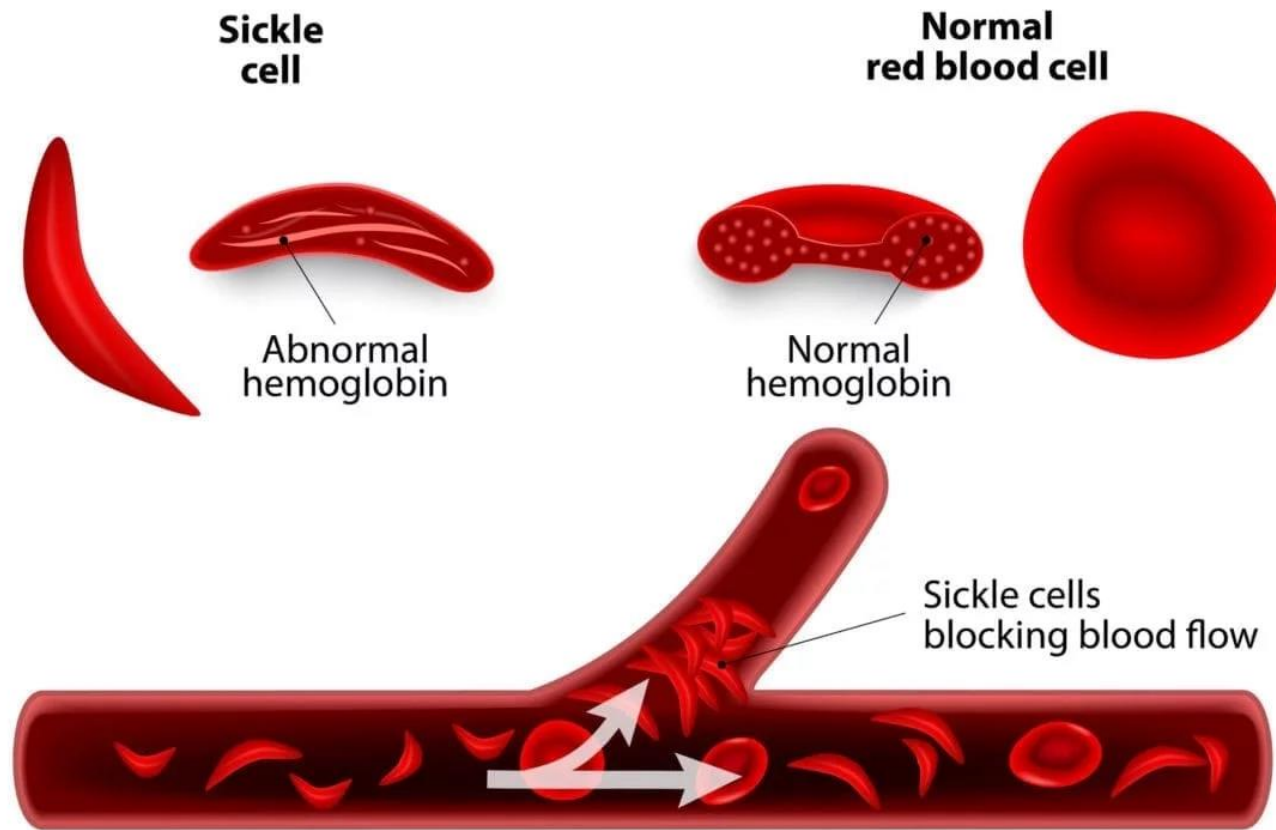


- Sickle cell disease (SCD) is a term used to describe a group of symptomatic hemoglobinopathies that have in common sickle cell formation and the associated crises.
- Patients with SCD are either homozygous for HbS (SS) or are compound heterozygotes expressing Hb S in combination with another Hb β chain mutation like Hb C or β -thalassemia.
- SCDs are the most common form of hemoglobinopathy, with Hb SS and the variants Hb SC and Hb S- β -thalassemia (HbS- β -thal) occurring most frequently.

Pathophysiology

- Pathophysiology is how a disorder translates into clinical symptoms.
- Glu (net charge: -1) → Val (net charge: 0) → change in charge of (+1).
- In the case of sickle cell mutation, the amino acid substitution results in Hb polymerization, leading to the formation of long Hb tactoids or long crystals that stretch the RBC membrane and produce the characteristic crescent moon or sickle cell shape.
- This change in charge affects the electrophoretic mobility of the Hb.

ANEMIA



- The sickling process is dependent on the degree of oxygenation, pH, and dehydration of the patient.
- Decreases in oxygenation and pH, as well as dehydration, promote sickling.
- In a homozygous state (HbSS), the sickling process begins when oxygen saturation decreases to less than 85%.
- In a heterozygous state (HbAS), the sickling process doesn't occur unless the oxygen saturation decreases to less than 40%.

Clinical features

- Can vary from no symptoms to a potentially lethal state with a variety of symptoms.
- Are not evident until the second half of the first year of life.
- RBCs containing Hb S become susceptible to hemolysis, and progressive hemolytic anemia and splenomegaly may become evident.
- Many individuals with SCD undergo episodes of recurring pain termed crises.

BOX 27-3 Clinical Features of Sickle Cell Disease

I. Vasoocclusion

A. Causes:

Acidosis

Hypoxia

Dehydration

Infection

Fever

Extreme cold

B. Clinical manifestations

1. Bones:

Pain

Hand-foot dactylitis

Infection (osteomyelitis)

2. Lungs:

Pneumonia

Acute chest syndrome

3. Liver:

Hepatomegaly

Jaundice

4. Spleen:

Sequestration splenomegaly

Autosplenectomy

5. Penis:

Priapism

6. Eyes:

Retinal hemorrhage

7. Central nervous system

8. Urinary tract:

Renal papillary necrosis

9. Leg ulcers

II. Bacterial infections

A. Sepsis

B. Pneumonia

C. Osteomyelitis

III. Hematologic defects

A. Chronic hemolytic anemia

B. Megaloblastic episodes

C. Aplastic episodes

IV. Cardiac defects

A. Enlarged heart

B. Heart murmurs

V. Other clinical features

A. Stunted growth

B. High-risk pregnancy

Laboratory diagnosis

- The disease is diagnosed early at about 6 months of age when Hb F is replaced with Hb S.
- The following laboratory tests should be performed to diagnose sickle cell disease:
 - 1) CBC
 - 2) Retics count
 - 3) Peripheral blood smear
 - 4) Solubility test
 - 5) Hb electrophoresis, HPLC, or capillary electrophoresis.

- Sickle cell anemia (HbSS): is a genetic or inherited chronic hemolytic anemia characterized by the two β chains are mutated ($\alpha_2 \beta_2$ 6Glu→Val).
- The disease is diagnosed early at about 6 months of age when Hb F is replaced with Hb S.
- Hb:↓ (6-8g/dL), Hct, RBCs:↓
- MCV, MCH: normal, Retics:↑(hemolytic crisis, 10-25%), ↓(a plastic crisis).
- RDW:↑
- Moderate leukocytosis with neutrophilia.
- Thrombocytosis is usually present.

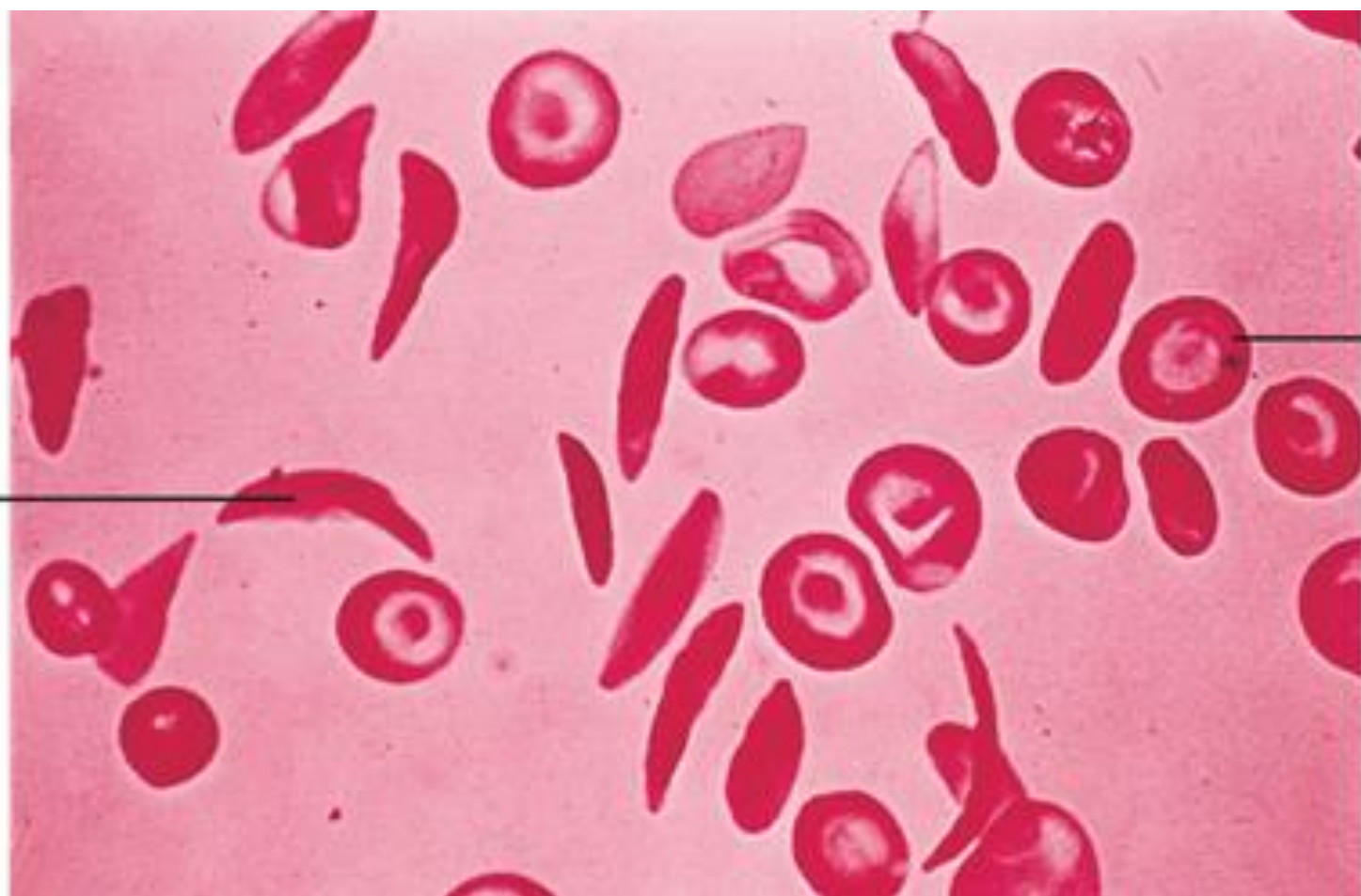
➤ **Blood film:**

- Normochromic normocytic RBCs.
- Sick cells (long curved pointed cells).
- Marked anisopoikilocytosis.
- Target cell.
- Nucleated RBCs.
- Basophilic stippling.
- Howell-Jolly (DNA remnant) bodies.
- Moderate to marked polychromasia (reticulocytosis).

➤ **BM: erythroid hyperplasia except during an aplastic crisis.**

➤ Presence of sickle cell and target cell is the hallmark of SCD.

Sickled
cell

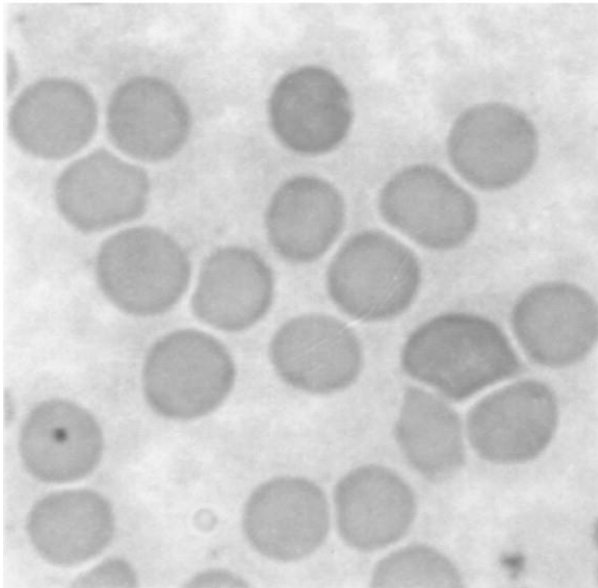


Normal
cell

Sickling Test

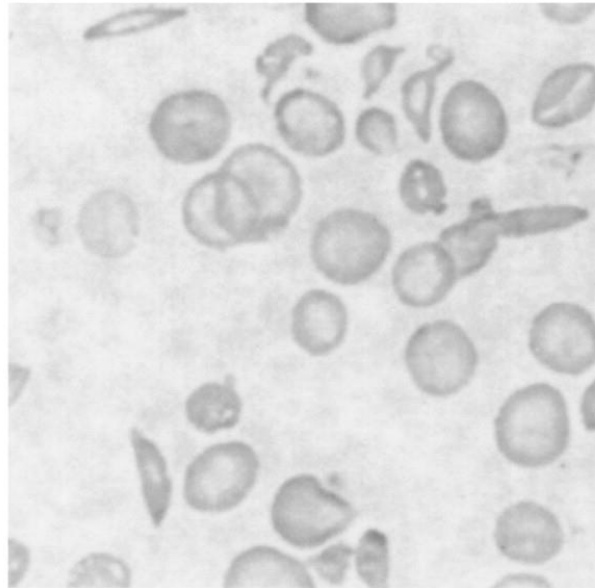
Sodium metabisulfate will cause the cells to sickle by deoxygenating the blood (seen under the microscope).

normal

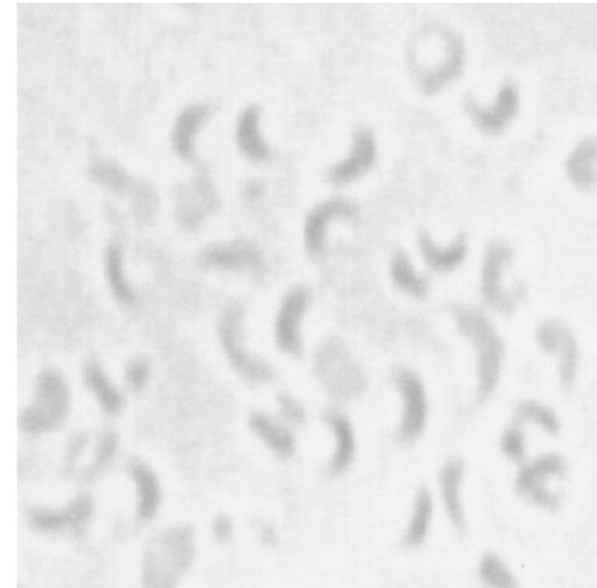


sickle cell

oxygenated



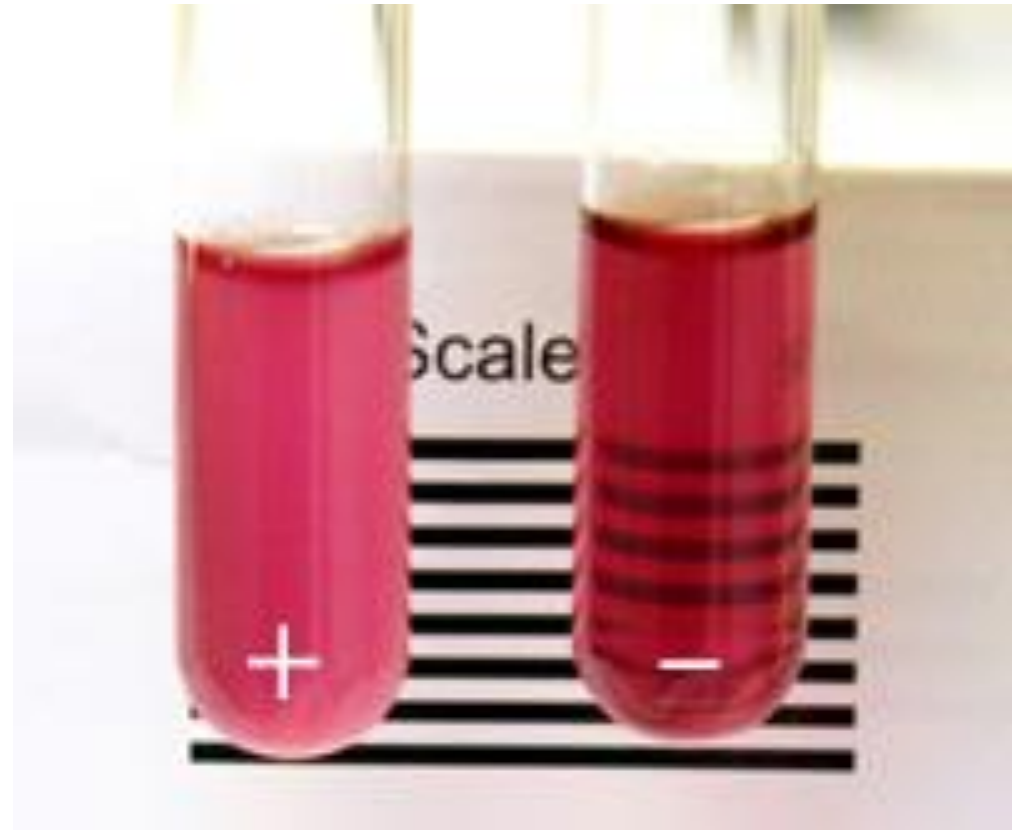
deoxygenated



Solubility test

Solubility test using a concentrated phosphate buffer, a hemolyzing, and sodiumdithionate.

- Mixture of Hb S in a reducing solution.
- Gives a turbid appearance.
- Normal HB gives a clear solution.



- The two screening tests do not distinguish between sickle trait and different forms of sickle cell disease.
- 1) Alkaline Hb electrophoresis on cellulose acetate at pH 8-8.4 is the common first step in the definitive diagnosis of hemoglobinopathy.
 - Hb S, D and G migrate with each other.
 - Hb C, E, O, and A2 migrate with each other.
- 2) Acidic Hb electrophoresis on citrate agar at pH 6-6.2 is used as a confirmatory test to help in distinguishing and differentiating between the same migrated Hbs.
 - Sickle cell anemia: Hb S: >80%, HbF: 1-20%, Hb A: absent

- Sickle cell trait (Hb AS): is a genetic or inherited asymptomatic disorder characterized by one β chain being mutated ($\alpha_2 \beta_2$ 6Glu→Val) and the other one being normal.
- Normal RBC morphology, except few target cells.
- Normal WBCs and platelets.
- Positive solubility test.
- Hb A: >60%, Hb S: <40%.
- No treatment is required.

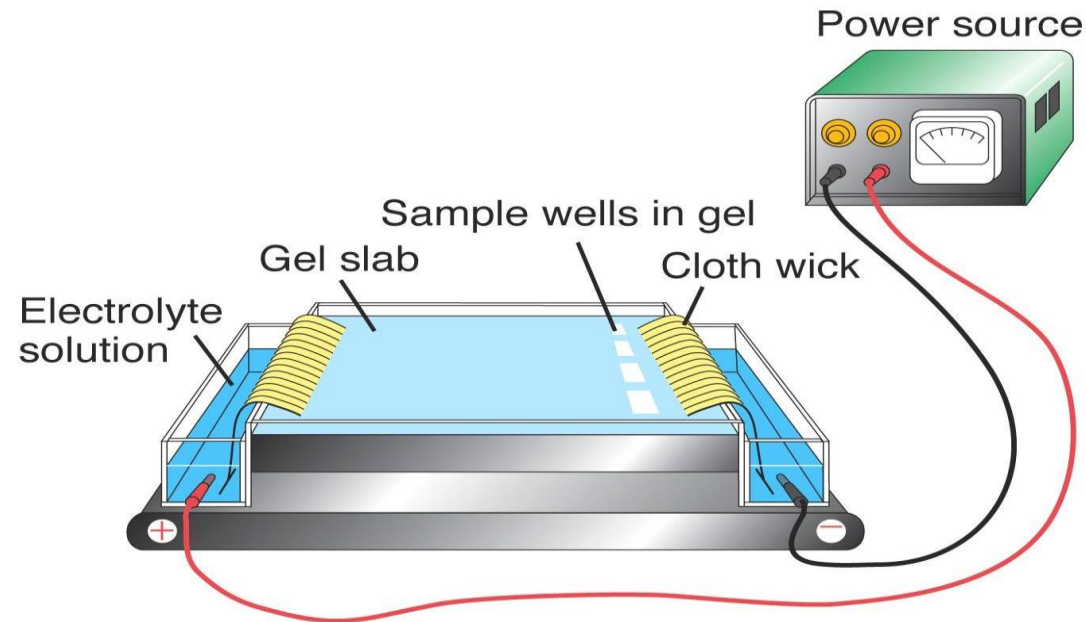
- During pregnancy, the mother is responsible for delivering oxygen to the baby and removing CO₂. For this to happen, maternal and fetal circulations come into very close proximity in the placenta.
- It is important that this system preferentially moves oxygen into the fetal circulation and CO₂ into the maternal circulation. This would not happen if both circulations had the same oxygen transporter with the same binding characteristics as the gases would bind equally between the two circulations.
- To overcome this, mammals have evolved a unique form of hemoglobin for fetuses, known as fetal hemoglobin (Hb F) consisting of two alpha and two gamma subunits ($\alpha_2 \gamma_2$).
- This allows Hb F to bind oxygen with a higher affinity than adult hemoglobin (Hb A), allowing extraction of oxygen from maternal circulation.
- This results in a left-shifted hemoglobin-oxygen dissociation curve compared to adult hemoglobin.
- This means oxygen loading occurs at lower oxygen concentrations.

Hemoglobin Electrophoresis

- Electrophoresis is a means of separating hemoglobins. It depends on the migration of the hemoglobin molecules dissolved in a buffer on, or in, a supporting medium when an electric current is passed through them.
- Hemoglobin electrophoresis is performed to find abnormal forms of hemoglobin (hemoglobinopathy).
- Healthy adults only have significant levels of HbA and HbA2.
- Some people may also have small amounts of HbF (which is the main type of hemoglobin in an unborn baby's body). Certain diseases are associated with high HbF levels (when HbF is more than 2% of the total hemoglobin).

Principle of Hb electrophoresis (cellulose acetate gel)

- In an alkaline pH (8.2-8.6) Hb is a negatively charged molecule and will migrate toward the anode (+).
- The various Hbs move at different rates depending on their net negative charge, which in turn is controlled by the composition (amino acids) of the Hb molecule (globin chain).

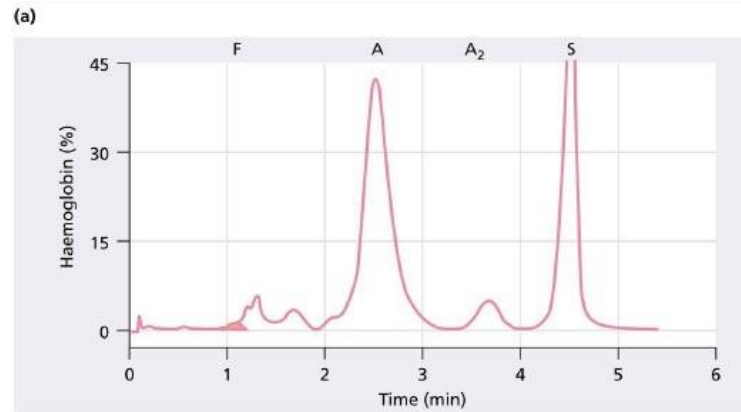
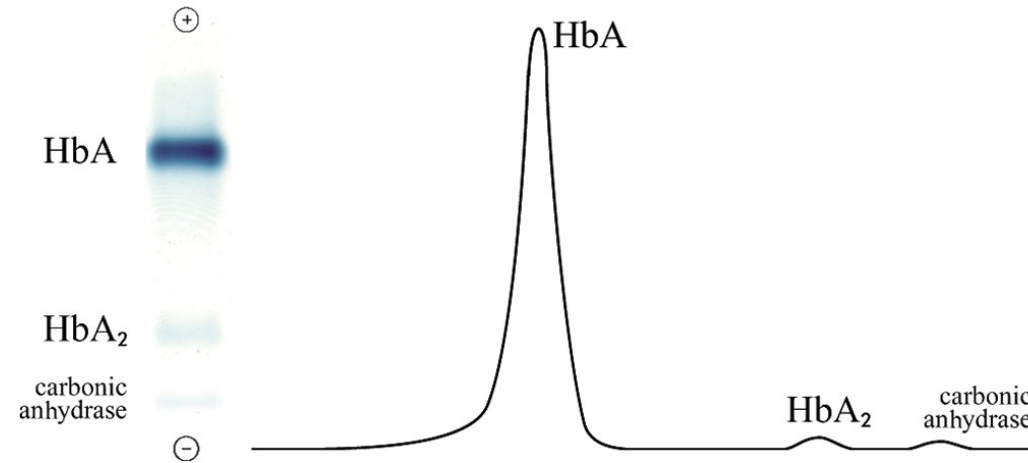


- The red cell hemolysate (red blood cell membranes are destroyed to free the Hb molecules for testing) is placed in a cellulose acetate membrane, which is positioned in an electrophoresis tray with the inoculated hemolysate near the cathode (-).
- One end of the cellulose acetate strip is immersed in the buffer (pH 8.2-8.6) on the cathode side and the other end is placed in the buffer on the anode (+) side. An electric current of a specific voltage is allowed to run for a timed period.
- During electrophoresis, the Hb molecules migrate toward the anode because of their negative charge. The difference in the net charge of the Hb molecule determines its mobility and manifests itself by the speed with which it migrates to the positive pole.

- The cellulose acetate membrane is then stained to color the proteins (Hbs). By noting the distance each Hb has migrated and comparing this distance with the migration distance of known controls, the types of hemoglobins may be identified.
- Examples of the fast Hbs are Hb Bart's and the two fastest variants Hb H and I, while Hb C is the slowest common Hb.

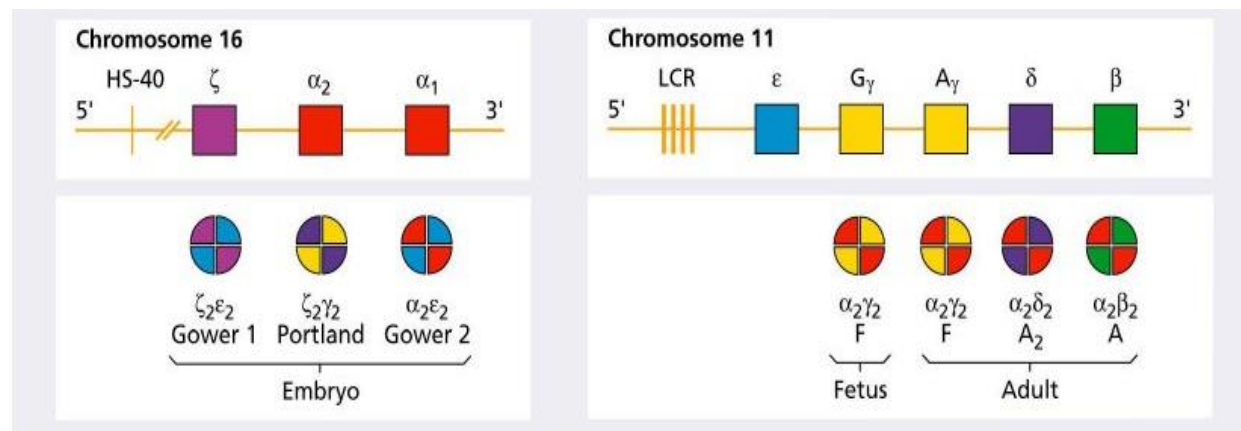
Normal Hb electrophoresis and result expression on a graph

(the area under the graph represent the concentration)



(b)

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Adult Hb:

Hb A: 95% to 98%

Hb A₂: 2% to 3%

Hb F: 0.8% to 2%

Hb S: 0%

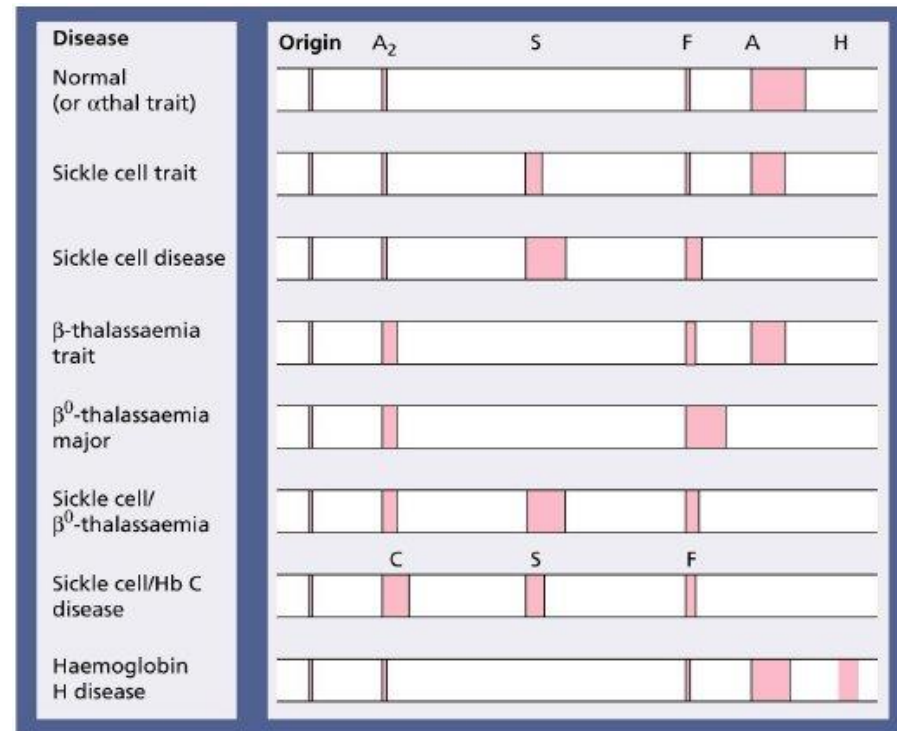
Hb C: 0%

Fetal Hb:

Hb F (newborn): 50% to 80%

Hb F (6 months): 8%

Hb F (over 6 months): 1% to 2%



Diagnosis of Hb genetic abnormalities by electrophoresis.

Adult Hb:

Hb A: 95% to 98%

Hb A2: 2% to 3%

Hb F: 0.8% to 2%

Hb S: 0%

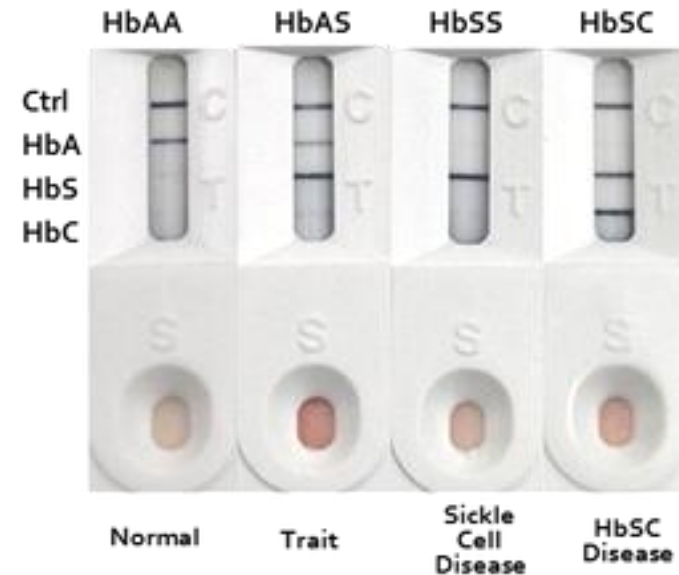
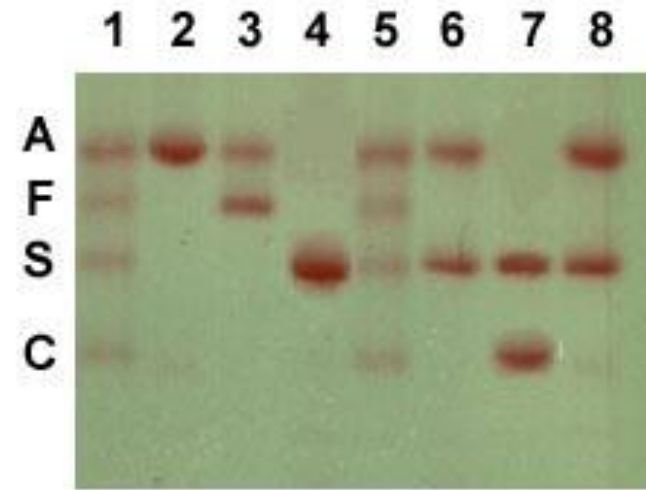
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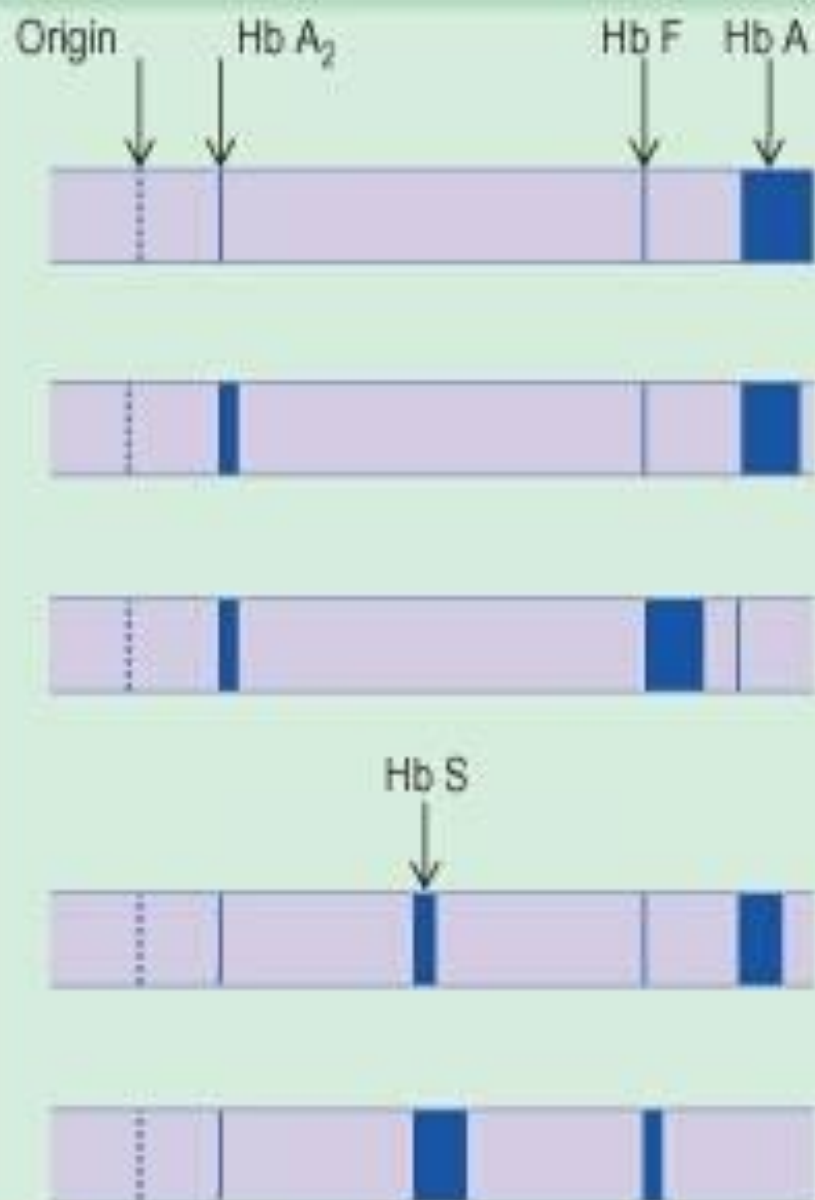
Normal

β -Thalassaemia
trait

β -Thalassaemia
major

Sickle cell
trait

Sickle cell
anaemia



Treatment

- Supportive therapy: adequate hydration, prophylactic vitamin therapy (folic acid), avoidance of low-oxygen environment, analgesics for pain, and antibiotics (penicillin).
- Periodic blood transfusion to prevent complications of SCD.
- Blood exchange transfusion: is the treatment of choice for severe vaso-occlusive crisis attacks and acute chest syndrome and aims to lower Hb S to less than 30%.
- BM or hematopoietic stem cell transplantation.
- Hydroxy urea therapy to increase the Hb F and decrease Hb S in the RBC) ↑Hb F: ↓severity).

Table 2.4 Classification of anaemia.

Microcytic, hypochromic	Normocytic, normochromic	Macrocytic
MCV <80 fL MCH <27 pg	MCV 80–95 fL MCH ≥27 pg	MCV >95 fL
Iron deficiency Thalassaemia Anaemia of chronic disease (some cases) Lead poisoning Sideroblastic anaemia (some cases)	Many haemolytic anaemias Anaemia of chronic disease (some cases) After acute blood loss Renal disease Mixed deficiencies Bone marrow failure (e.g. post-chemotherapy, infiltration by carcinoma, etc.)	Megaloblastic: vitamin B ₁₂ or folate deficiency Non-megaloblastic: alcohol, liver disease, myelodysplasia, aplastic anaemia, etc. (see Table 5.10)
MCH, mean corpuscular haemoglobin; MCV, mean corpuscular volume.		